Miloš Medić

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Education

BSc **Faculty of Technical Sciences (FTN), University of Novi Sad**, Electrical Engineering and Computer Science — Oct 2021 – present

- **GPA: 9.57/10**, **Top 3% of graduating class**
- Module: **Applied Computer Science and Informatics**
- All coursework completed; thesis in progress, expected graduation: **Sep 2026**
- Relevant coursework: **Security Engineering**, **Distributed Systems**, **Computer Architecture**, Microservices Architecture

Experience

Faculty of Technical Sciences (FTN), University of Novi Sad, Course Demonstrator, Computer Architecture — Novi Sad, Serbia
Oct 2022 – Feb 2023
5 months

- Provided hands-on debugging and troubleshooting support for students during **computer architecture laboratory sessions**

Projects

PKI and Zero-Knowledge Password Vault — Jan 2026 – Feb 2026

- Designed and implemented a formal **Public Key Infrastructure (PKI)** framework conforming to standard certificate validation paths, incorporating Root, Intermediate, and End-Entity certificate authorities with **CRL/OCSP revocation mechanisms**
- Developed a **zero-knowledge cryptographic model** utilizing the Web Crypto API for client-side **RSA-OAEP** encryption, ensuring strict data privacy protocols where private keys are never exposed to the persistence layer
- Evaluated and mitigated system vulnerabilities through role-based access control (RBAC), multi-factor authentication (2FA), and secure asymmetric key exchange for cross-user data sharing, resilient against injection and XSS vectors

Distributed Microservices Tourism Platform — Nov 2025 – Dec 2025

- Analyzed and implemented a polyglot microservices architecture (Go/Java) to study service isolation and communication patterns via containerized environments and custom API gateways
- Applied the **SAGA orchestration pattern** to evaluate data consistency and distributed transaction management across heterogeneous database paradigms, balancing relational (PostgreSQL), document (MongoDB), and graph (Neo4j) engines
- Integrated a full **observability and metrics stack** using Prometheus and Grafana for empirical analysis of distributed tracing, log aggregation, and real-time container resource allocation

Desert Oasis, Real-Time 2D OpenGL Scene — Oct 2024 – Nov 2024

- Engineered a low-level 2D graphics execution pipeline from scratch using **C++ and raw OpenGL**, demonstrating explicit control over rendering state machines and geometric coordinate transformations
- Implemented a mathematical model for procedural environmental simulation, computing dynamic sky gradients and celestial trajectories (sun/moon arcs) mapped directly to time-based state transitions
- Designed a real-time **event-driven interaction framework** managing user input constraints, state-driven animations, and deterministic execution sequences for complex graphics scenes

Skills

Low-Level & Systems: C, **C++**, **x86 Assembly**, OpenGL
Programming Languages: Java, Python, Go, C#, JavaScript, TypeScript, Julia, MATLAB, HTML/CSS
Frameworks & Platforms: Spring Boot, .NET, React, Angular, Vue, Next.js, Node.js, Android (Java)
Databases: PostgreSQL, MongoDB, MySQL, SQLite, Firebase, Neo4j
Infrastructure & DevOps: Docker, Docker Compose, RabbitMQ, Prometheus, Grafana
Security & Protocols: X.509/PKI, JWT, Auth0, Keycloak, HTTPS/TLS, CRL/OCSP, Web Crypto API, REST, gRPC
Tools: Git, GitHub, GitLab, VS Code, IntelliJ IDEA, Postman, Swagger